

Dominique Nshimiyimana

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🔍 Google Scholar  ORCID

Employment History

- 2022 – Present  **Wissenschaftlicher Mitarbeiter, RPTU Kaiserslautern-Landau in Embedded Intelligence Group**
Research on sustainable embedded intelligence, focusing on resource-efficient machine learning, wearable sensing, and human activity recognition.
- 2015 – 2022  **Student Research Assistant (HiWi), DFKI.**
Contributed to the development of software tools for data and workflow management, including database applications, data processing, and software development.
- 2014 – 2015  **Robotik Club, TU Kaiserslautern.**
Member of the student team participating in the International *Field Robot Event* (2014–2015), developing autonomous robotic systems.

Education

- 2019 – 2022  **M.Sc. in Computer Science, TU Kaiserslautern.**
Thesis: *Deep Learning Based Semantic Segmentation to Improve Spectator Safety in WRC Races.*
- 2014 – 2019  **B.Sc. in Computer Science, TU Kaiserslautern.**
Thesis: *Semantic Segmentation for 3D Reconstruction with Spherical Images.*

Research Publications

Journal Articles

- 1 D. Geissler, D. Nshimiyimana, V. F. Rey, S. Suh, B. Zhou, and P. Lukowicz, “Beyond confusion: A fine-grained dialectical examination of human activity recognition benchmark datasets,” *arXiv preprint arXiv:2412.09037*, 2024.

Conference Proceedings

- 1 D. Nshimiyimana, V. F. Rey, S. Suh, B. Zhou, and P. Lukowicz, “Pim: Physics-informed multi-task pre-training for improving inertial sensor-based human activity recognition,” in *2025 International Conference on Activity and Behavior Computing (ABC)*, 2025, pp. 1–11.  DOI: 10.1109/ABC64332.2025.11118365.
- 2 M. Liu, D. Geißler, D. Nshimiyimana, S. Bian, B. Zhou, and P. Lukowicz, “Initial investigation of kolmogorov-arnold networks (kans) as feature extractors for imu based human activity recognition,” in *Companion of the 2024 on ACM International Joint Conference on Pervasive and Ubiquitous Computing*, ser. UbiComp ’24, Melbourne VIC, Australia: Association for Computing Machinery, 2024, pp. 500–506, ISBN: 9798400710582.  DOI: 10.1145/3675094.3678446.
- 3 J. Sorysz, L. Krupp, D. Nshimiyimana, and P. Lukowicz, “Beyond the pocket: Preparing a study on the variations of wearable device location,” in *Proceedings of the 13th International Conference on the Internet of Things*, ser. IoT ’23, Nagoya, Japan: Association for Computing Machinery, 2024, pp. 228–231, ISBN: 9798400708541.  DOI: 10.1145/3627050.3631575.

- 4 V. Fortes Rey, D. Nshimiyimana, and P. Lukowicz, “Don’t freeze: Finetune encoders for better self-supervised har,” in *Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing*, ser. UbiComp/ISWC ’23 Adjunct, Cancun, Quintana Roo, Mexico: Association for Computing Machinery, 2023, pp. 195–196, ISBN: 9798400702006. [DOI: 10.1145/3594739.3610790](#).

Books and Chapters

- 1 D. Nshimiyimana, V. Fortes Rey, and P. Lukowicz, “Contrastive left-right wearable sensors (imus) consistency matching for HAR,” in *Activity, Behavior, and Healthcare Computing*, CRC Press, 2025, pp. 88–104. [DOI: 10.1201/9781032648422-7](#).

Skills

Languages	English (professional proficiency) and German (basic proficiency)
Programming	Python (PyTorch, NumPy, SciPy, scikit-learn, OpenCV), Java, and $\text{\LaTeX}$.
Tools	Git, Linux, SQL databases, and scientific computing tools.